EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics

Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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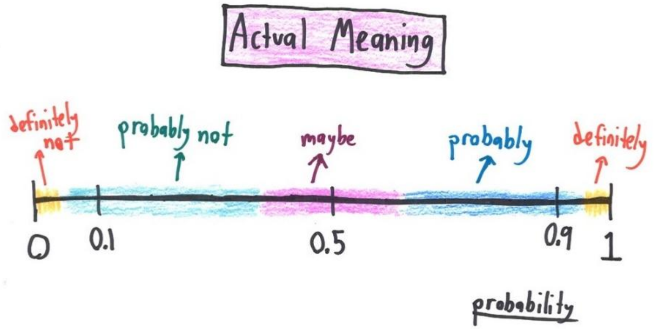
Week 5 - Lecture No 07 – 15th September 2025



Announcements!

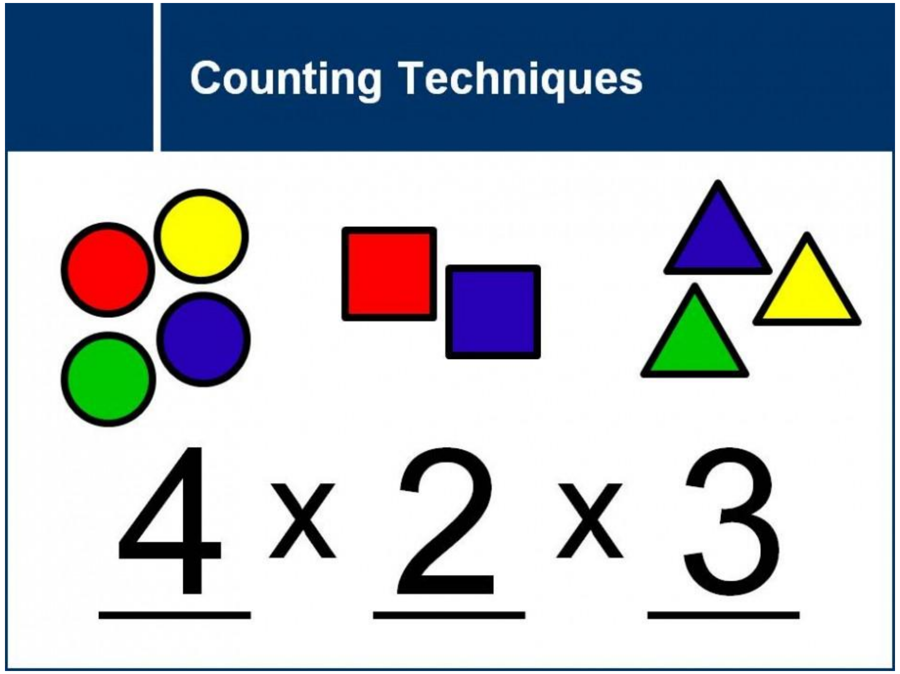
1. Lectures up to No 06 posted
2. Additional lecture - September 20, 2025 (Saturday, same time & place) in-lieu of the class missed on Wednesday August 20, 2025
3. An added lecture will be posted in a asynchronous mode in lieu of September 08, 2025 class
4. Quiz No 02 – September 17/19/20/21, 2025

Agenda for today

1. Complete Unit 2 - Conditional Probability and Bayes' Rule
2. Unit 3 - Counting

Unit 3: Counting

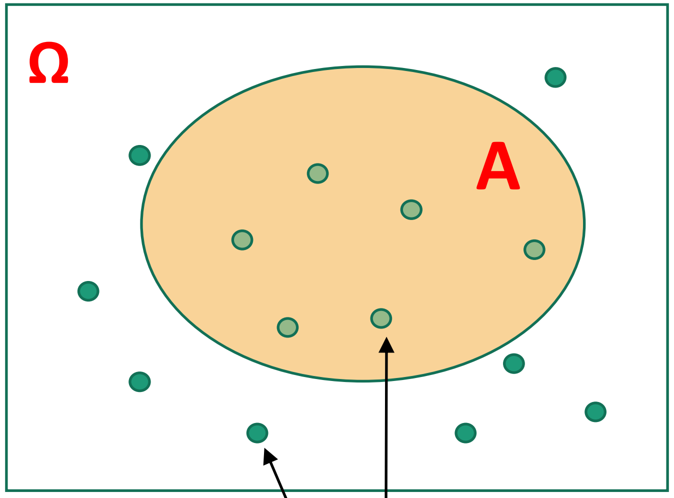
- Permutations
- Combinations



Counting – Why?

Discrete Uniform Law:

- Assume Ω consists of n equally likely elements
- Assume A consists of k elements



$$prob = \frac{1}{n}$$

Then: $\mathbf{P}(A) = \dfrac{\text{number of elements of } A}{\text{number of elements of } \Omega} = \dfrac{k}{n}$

Practice Problem 1

"A certain class of 20 HU students are asked to undertake a class project comprising of two students. A geek HU programmer decides to write a Python program that lists all possible groups (N_1 = Number of possible groups) that can possibly be formed. She takes her program to a new level and based on some crude AI also predicts the group experience U_1 – based on student profiles and maps this on a scale of 3 (1-crappy, 2-ok, 3-excellent)"

What is N_1?

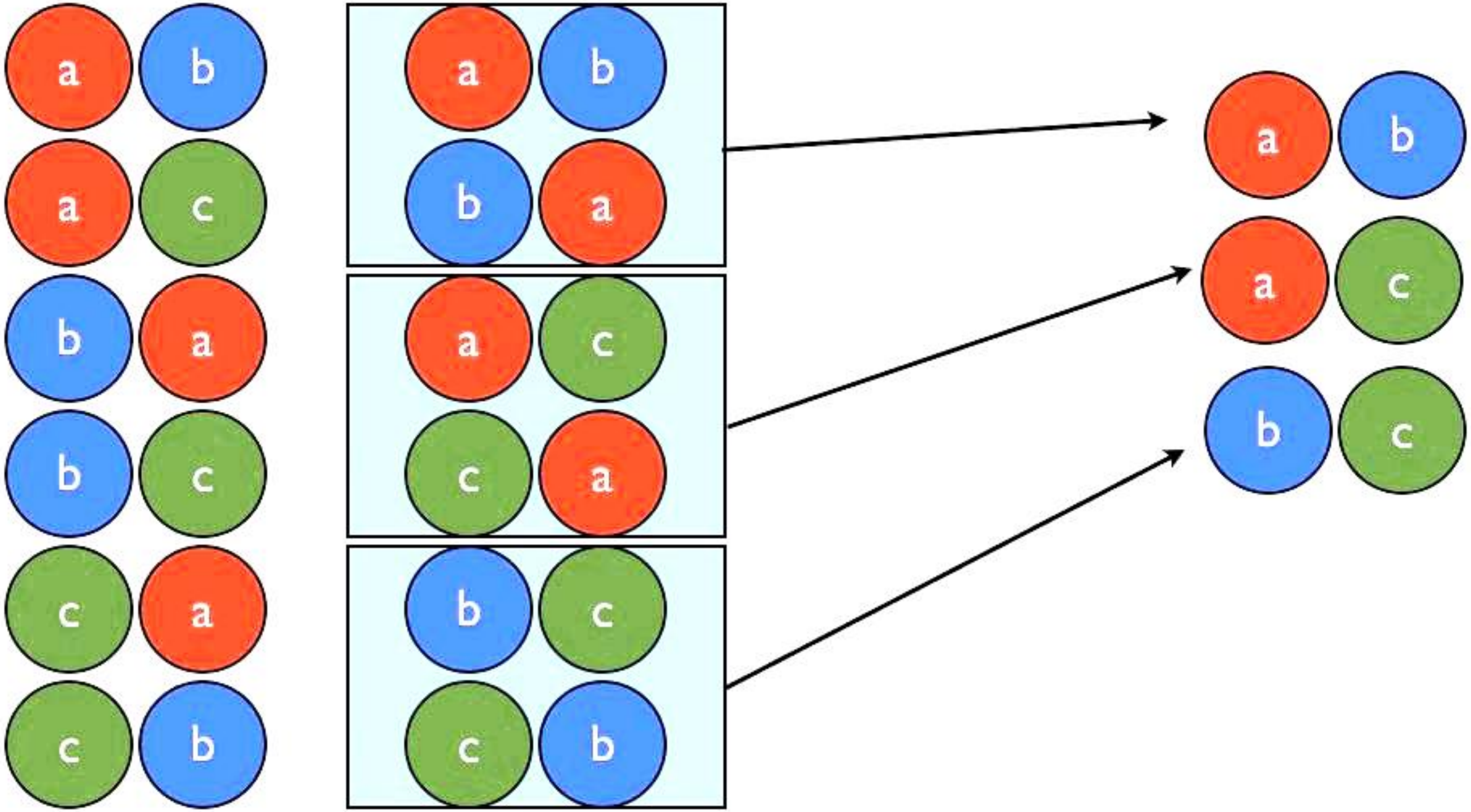
"Based on a very poor feedback with regards to the class project exercise, in the next offering the instructor now asks each group to select a group lead & a member within each group. Assuming that this class also has 20 HU students, the number of possible group curations, N_2, are now mapped (using the same crappy AI) to predict the user experience U_2"

What is N_2?

After group activity – Discuss U_1 versus U_2

Counting

$_3\mathbf{P}_2$ $\quad$ $_3\mathbf{P}_2 = 6$ $\quad$ $_3\mathbf{C}_2$



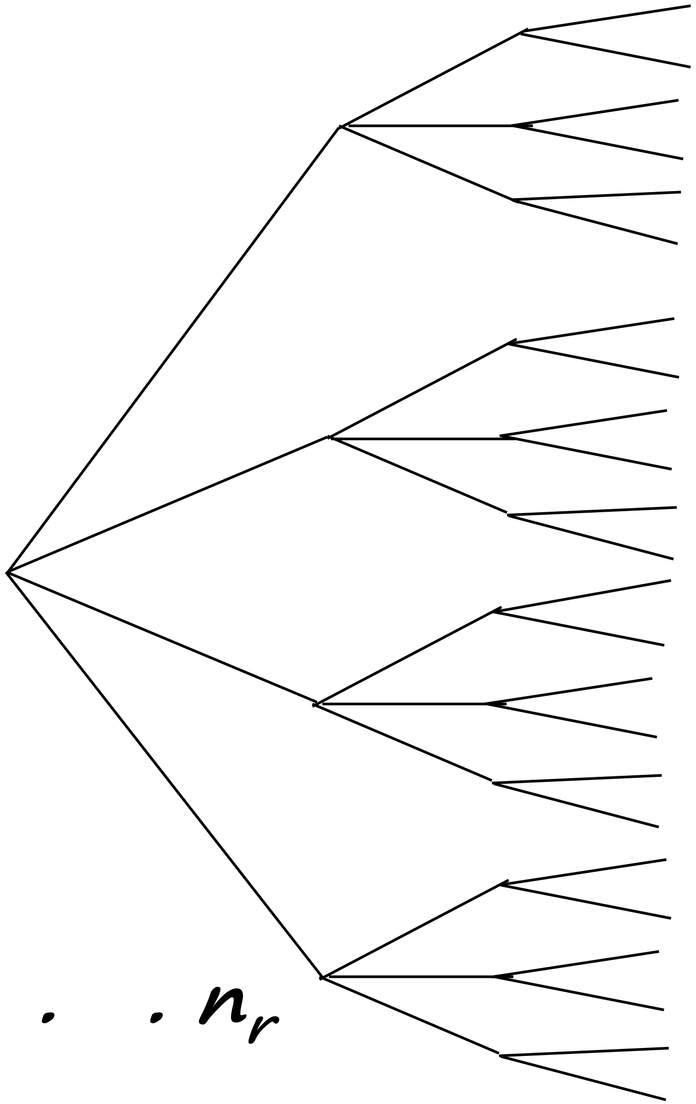
Basic Counting Principle

4 12 24 = 4 x 3 x 2

Example:

- 4 shirts.
- 3 ties.
- 2 jackets.

Number of possible attires?



$r = 3$

$n_1 = 4$

$n_2 = 3$

$n_3 = 2$

- r stages
- n_i choices at stage i

➡ Number of choices is $n_1 \cdot n_2 \cdot n_3 \cdot \;\cdot\; \cdot\; \cdot n_r$

Basic Counting Principle (ctd)

Example:

- **Number of license plates with 2 letters followed by 3 digits:**

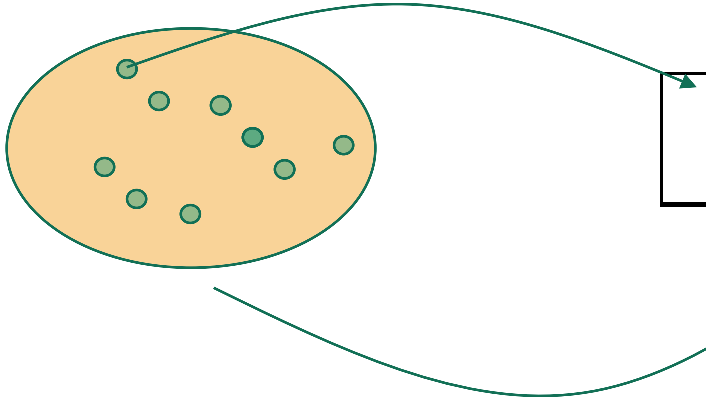
  ➡ $26 \cdot 26 \cdot 10 \cdot 10 \cdot 10$

  - What if repetition is prohibited?

    ➡ $26 \cdot 25 \cdot 10 \cdot 9 \cdot 8$

- **Permutations**: *Number of ways of ordering n elements*

  ➡ $= n \cdot (n-1) \cdot \ldots \cdot 1 = n!$

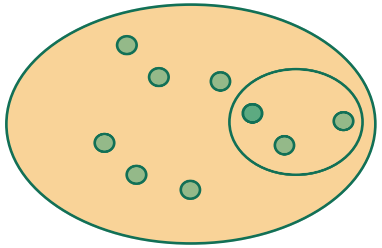


n	$n-1$	$n-2$	.	.		1

- **Number of distinct subsets of {1, … , n}:** $= 2^n$

Remember this!

?



2	2	2	.	.		2

Example

- Find the probability that:
  six rolls of a (six sided) die all give different numbers.
  (Assume all outcomes equally likely.)

A

➡ *Typical outcome:* $P(2, 3, 4, 5, 6, 2) = \dfrac{1}{6^6}$

➡ *Typical element of A:* $(2, 3, 4, 1, 6, 5) = 6!$

➡ $$P(A) = \frac{\#\ in\ A}{\#\ possible\ outcomes} = \frac{6!}{6^6}$$

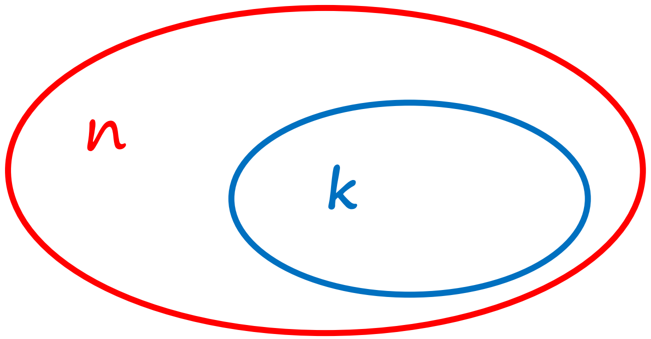
Combinations

Definition: $\binom{n}{k}$ Number of *k*-element subsets of a given *n*-element set

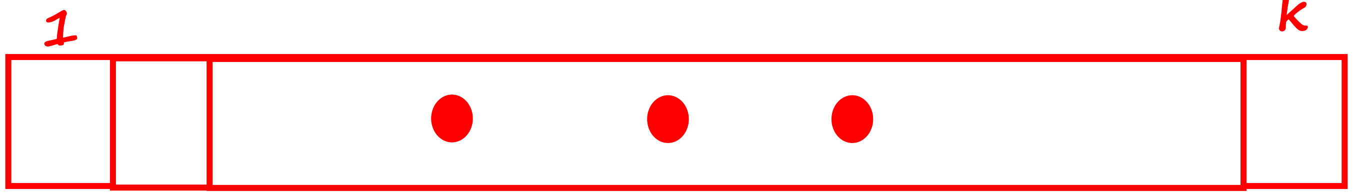
$$= \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

➡ $n = 0, 1, 2, \dots$

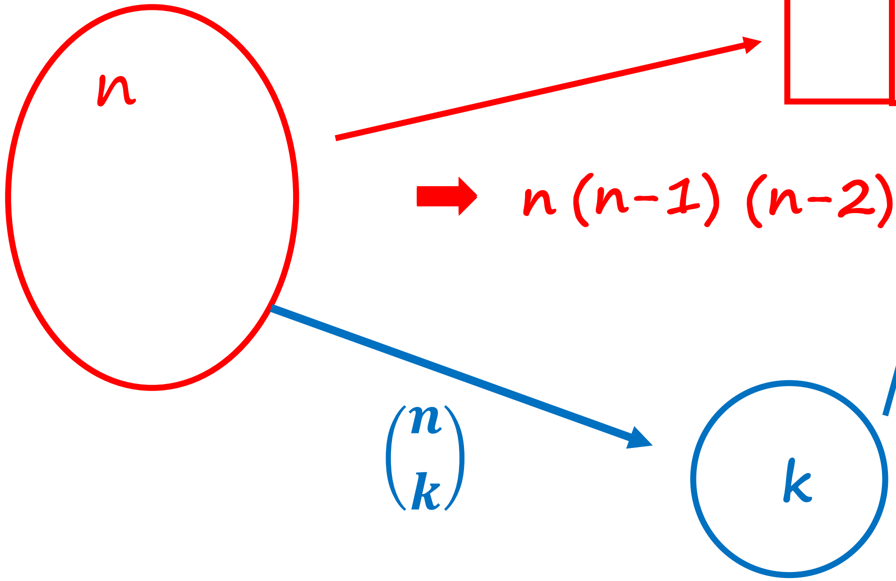
➡ $k = 0, 1, 2, \dots, m$



- Two ways of constructing an **ordered** sequence of *k* **distinct** items:
  - Option 1 - Choose the ***k*** items one at a time.
  - Option 2 - First Choose ***k*** items, then order the ***k*** them.



➡ $n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!}$



$k!$

$\binom{n}{k}k!$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\binom{n}{n} = 1 \qquad \frac{n!}{n!\,0!} \qquad 0! = 1 \text{ convention}$$

$$\binom{n}{0} = \frac{n!}{0!\,n!} = 1 \qquad \emptyset$$

$$\sum_{k=0}^{n} \binom{n}{k} = \binom{n}{0} + \binom{n}{1} + \cdots \binom{n}{n} = \#\ all\ subsets = 2^n$$